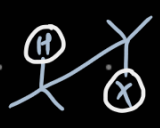
ELIMINATION

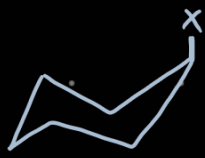
REACTIONS MECHANISMS

	E_1	E_2
1. Rate	$R = K[subs]^1$	$R = K[subs]^1[B]^1$
2. Molecularity	1	2
3. Order	1	2
4. Favoured Subst.	$3^\circ > 2^\circ > 1^\circ > CH_3X$	$3^\circ > 2^\circ > 1^\circ > CH_3X$
5. Feasibility	Stability of $C^\oplus$ + (weak Nu) ↳ 1. Aromatic 2. Resonance 3. Hyperconjugation 4. Inductive	Strong base (charged) ↳ small → Saytzeff ↳ Bulky → Hoffman (S+H) * Isotope Effect ✓ * In 3° subs → Base like { $RO^\ominus$, $^\ominus OH$, $^\ominus NH_2$ } → E_2
	* Polar solvent	
6. RDS	Formation of $C^\oplus$ (first)	Single Step rxn.
7. Hybridisation	$sp^3 \rightarrow sp^2 \rightarrow sp^3$	$sp^3 \cdots sp^2$
8. Stereochemistry	–	Anti Elimination
9. To be paid attention	$C^\oplus$ rearrangement	Stereochemistry

+ Regioselective ⇒ favouring one isomer > other in product

+ Anti-periplanar : 

Elimination hone ke liye aisi conformation zaroori hai.



↳ L.G must be at axial.

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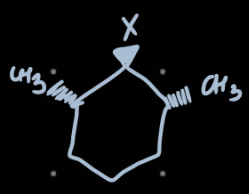
Trans = Agar X Axial ✓ ton CH_3 bhi Axial ✓ → 1-3 diAxial interaction

rate

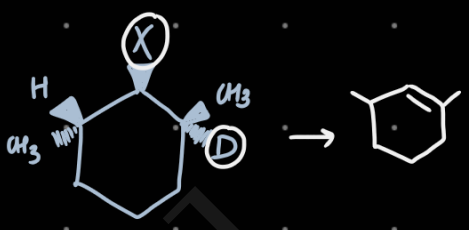


Cis ⇒ ek Axial toh dusra equitorial → no 1-3 diAxial inter. ✓ stable T.S

(Big groups like t-butyl can arrest ↳ No E_2 xxx)



→ XXX (Anti 'H' hona jaroori hai)

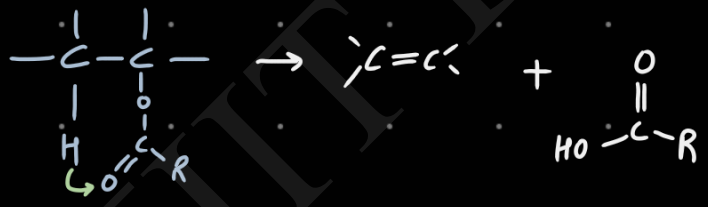


pehli cheez → Anti position
uske baad → H > D

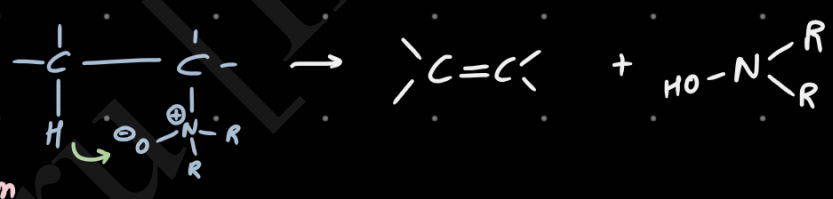
10. Pyrolytic Syn Elimination (E_i)

↳ intramol. → Base andar hi baitha hai!

① Esters

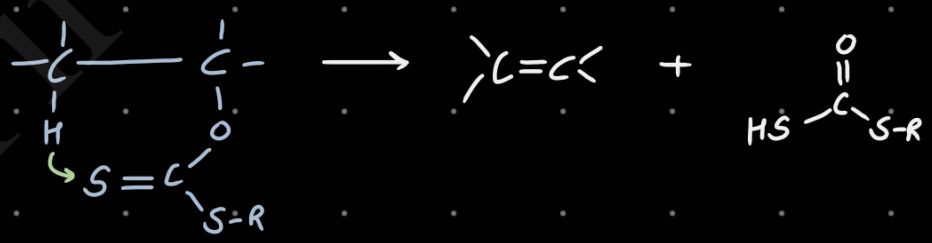


② Amine Oxides (tert. Amine ↳ Cope Elimination)



SYN!

③ Xanthates



★ R = K [subs]

↳ E_1 se alag ⇒ by stereochemistry ^ Syn hota hai

11. Dihalides

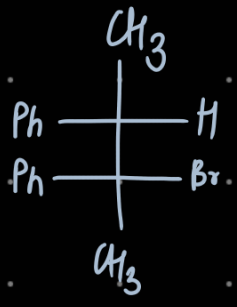


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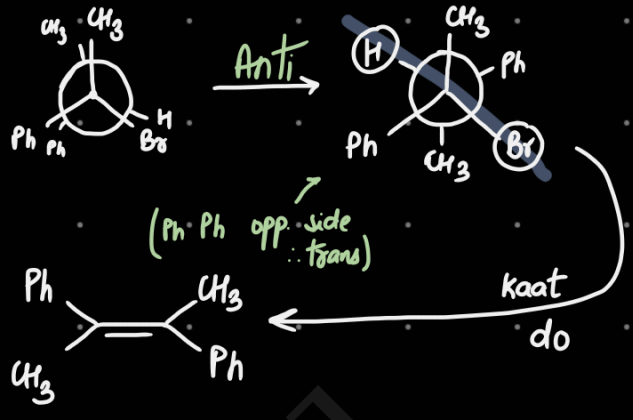
★ If both C → 1° or one is 1°, other 2° ⇒ SYN!!



12.



→ newman banao!

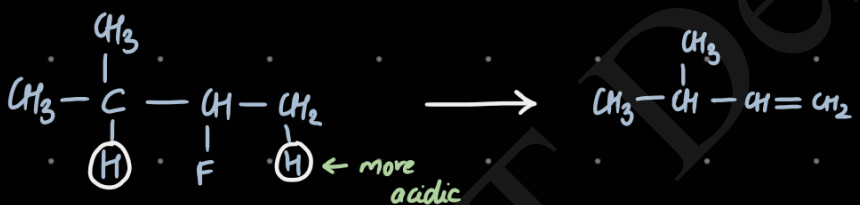


13. Poor LG

↳ isko nikaalne ke liye ⇒ $C^{\delta-}$ chaiye.

↳ jaise jaise LG ganda hota jaega ⇒ $C^{\delta-}$ ↑↑ ← "H" acidic ↑↑

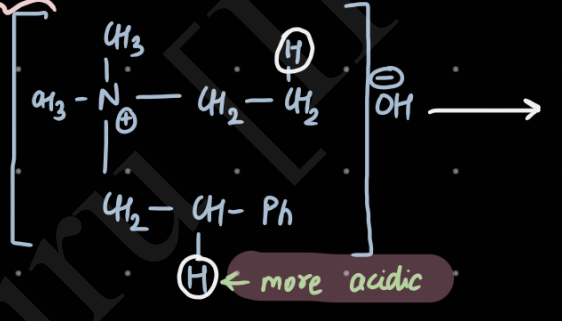
eg:



i°) Hoffman Elimination

(Quaternary ammonium salt)

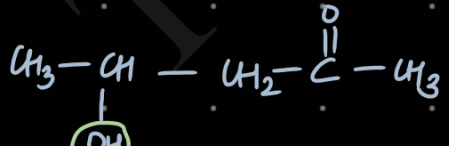
$R_3N \xrightarrow[AgOH]{R-X} R_4N^{\oplus}OH^{\ominus}$



$CH_2 = CH - Ph$

(also in $-\overset{\oplus}{S}(Me)(Me)$)

eg:



ii) E1cb

↳ pehle H^+ nikalega ⇒ $C^{\ominus}$ banega (C.B)

↳ stable by eWG.

★ Equil. in first step (fast)

Step 2 ⇒ L.G ka jaana → RDS

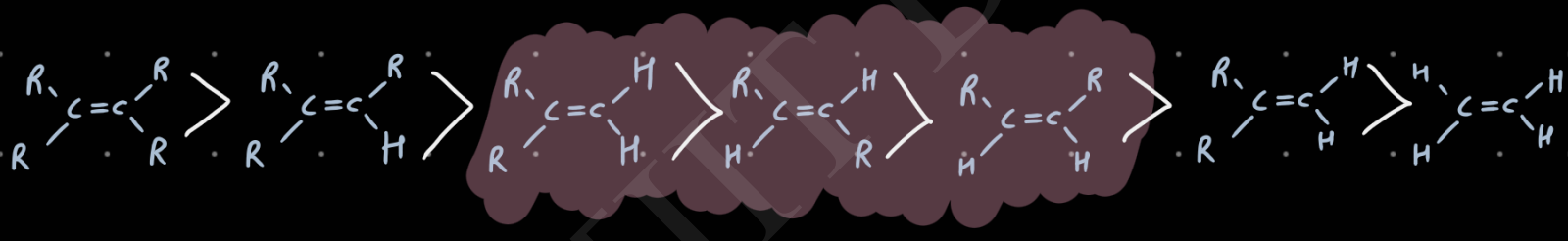
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★ Rate $\Rightarrow$ $K[C\cdot B] = K'[\text{Subs}][\text{Base}]$

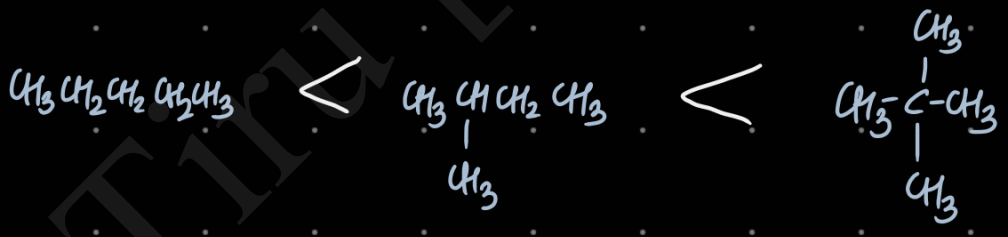
14.

	$E_{1cb}(R)$	$E_{1cb}(I)$
i. Step I	$\text{Subs} + B^{\ominus} \rightleftharpoons C\cdot B^{\ominus} + HB$ (fast)	$\text{Subs} + B^{\ominus} \longrightarrow C\cdot B^{\ominus} + HB$ (slow)
ii. Step II	$-\overset{\ominus}{C}-\overset{|}{\underset{|}{C}}- \longrightarrow >C=C< + X^{\ominus}$ (slow)	$-\overset{\ominus}{C}-\overset{|}{\underset{|}{C}}- \longrightarrow >C=C< + X^{\ominus}$ (fast)
iii. If solvent is EtOD	$-\overset{D}{\overset{|}{C}}-\overset{|}{\underset{|}{C}}-X$ is formed	XXX not formed

15. Alkene Stability



Alkane



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